UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN

WEDNESDAY, 8 MAY 2024 TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in Section A and **ONLY ONE (1)** question in Section B. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

Section A (Compulsory Questions)

Q1. (a) Discuss the differences between functional requirements and non-functional requirements used in system modeling. (4 marks)

(b) What is a project charter? (1 mark)

(c) A company is considering an investment in a new project that requires an initial investment of RM 10,000. The expected cash flows from the project over the next five years are as follows: RM 2,000 in Year 1, RM 4,000 in Year 2, RM 6,000 in Year 3, RM 8,000 in Year 4, and RM 10,000 in Year 5. The discount rate is 25%. Give the formulas:

$$PV = \frac{BC}{(1+r)^n}$$

where PV is present value per year, BC is the benefits or costs value, r is the discount rate, and n is nth years.

$$NPV_k = \sum_{t=1}^{k} \frac{BC_t}{(1+r)^t}$$

where NPV_k is accumulated kth years net present value, BC is the benefits or costs present value, r is the discount rate.

(i) Calculate the Net Present Value (NPV). (3 marks)

(ii) Calculate the Break-Even point. (3 marks)

(d) Briefly explain the **FIVE (5)** phases of the system development life cycle (SDLC). (10 marks)

(e) List **FOUR (4)** key factors used for selecting an appropriate requirements determination technique. (4 marks)

[Total : 25 marks]

Q2. (a) The course attendance management system allows staff to manage attendance and the students to check-in their attendance. Firstly, the staff are required to initiate an attendance session for the course to allow student check-in. During the attendance initiating process, the staff are required to fill in the information such as course code, teaching mode, venue, the start, and end time of the class. Upon the successful of the initiating process, a six-digit pin code will be generated, and students are required to use it for the check-in. At the end of the class, the attendance session will be automatically closed, and the student will no longer be able to check-in. Attendance is part of a course. Each student enrolls between one to five courses. Thus, only students enrolled in the course will be able to view the attendance and to check-in. Given the mentioned scenario, use the class diagram with invariants notation to draw the design of data structure with items listed below.

(i) Aggregation relationships between classes. (5 marks)

(ii) Association relationships between classes. (5 marks)

(iii) The usage of invariants and visibility. (3 marks)
[Hints: Combine the answer (i), (ii) and (iii) into a class diagram with invariants]

(b) What is architecture centric in term of object-oriented approach? List **THREE (3)** basic software architecture views used in architecture centric. (4 marks)

(c) Explain **THREE (3)** metrics used to evaluate an object-oriented system design in term of class design and method design. (6 marks)

(d) Explain the meaning of encapsulation in term of object-oriented concept.
(2 marks)
[Total : 25 marks]

Q3. (a) Undoubtedly, airport boarding system is the most critical component in an airport day-to-day operation. The boarding system has a few core functionalities. First, passengers that consist of child passenger, passenger with physical disabilities and tour guides should be able to perform individual check-in and security screening. Tour guides can choose to undergo group check-in, instead of individual check-in. For the passengers who bring along their luggage, they have to check-in their luggage. For the individual check-in, a passenger can either go for counter check-in or kiosk check-in. Given the mentioned scenario, use the use case diagram notation to draw the system functionalities with items listed below.

(i) Actor(s) who will interact with the system. (2 marks)

(ii) Use case that associated with actor(s) using association relationship(s). (3 marks)

(iii) Use case that associated with others use case(s) using either include or extend relationship(s). (2 marks)

(iv) Generalization relationship(s). (4 marks)

(v) System boundary. (1 mark)

[Hints: Combine the answer (i), (ii), (iii), (iv), and (v) into a use case diagram]

(b) List and explain **THREE (3)** data flow diagram (DFD) components. (6 marks)

(c) The electronic wallet (e-wallet) comprises a transferable amount and a non-transferable amount. Payment will be subtracted from the e-wallet only if the user possesses adequate funds. Initially, the payment amount will be reduced by the value of any applied voucher, if available. In cases where no voucher or the voucher value is insufficient to cover the payment, the non-transferable amount will be used if it can cover the payment. If the non-transferable amount is insufficient, the remaining balance, after deducting the non-transferable amount, will be subtracted from the transferable amount. Use a decision table to represent this logic. (7 marks)

[Total : 25 marks]

Section B (Choose any ONE Question)

Q4. (a) Tutors in an organization are assigned to teach courses according to the area of specialization and their availability by the course administrator. Every year, organizational staff is required to publish different courses calendar, maintain different courses calendar, and assigned tutors to the course respectively. In addition to that, course administrators in the organization will manage the course including course content, assign courses to the tutors, and specify course schedules. Given the mentioned scenario, use the window navigation diagram notation to illustrate the user interface design for the items listed below.

(i) Course administrator. (6 marks)

(ii) Organization staff. (8 marks)

(b) A layer represents an element of the software architecture of the evolving system. List **FIVE (5)** types of layers to be considered in design models. (5 marks)

(c) Compare and contrast the server-based architecture and client-server architecture in terms of security and ease of development. (6 marks)

[Total : 25 marks]

Q5. (a) Discuss the differences between high-level network diagram and low-level network diagram in terms of the system physical architecture design. Provide examples to illustrate both types of network diagram respectively to support your explanation. (6 marks)

(b) Upon user checkout of a redemption request, the system will concurrently deduct the user's points, generate an invoice, and send a request to the warehouse to prepare the selected item, if the user having sufficient points. Simultaneously, the system will send a notification message to the user via email, confirming the successful redemption process. Given the mentioned scenario, use the activity diagram notation to illustrate it for the items listed below.

(i) Step-by-step activities of the process. (2 marks)

(ii) Decision nodes used for the process. (2 marks)

(iii) Folk nodes and join node used for the process. (1 mark)

(iv) initial node and final-activity node usage. (1 mark)

[Hints: Combine the answer (i), (ii), (iii), and (iv) into an activity diagram]

Q5. (Continued)

(c) Discuss the normalization concept in term of the database design. (5 marks)

(d) Identify **FOUR (4)** types of the operational requirements used in design phase. (4 marks)

(e) Explain the differences between interface structure design and interface standard design of the user interface design process. (4 marks)

[Total : 25 marks]
